https://pubsonline.informs.org/journal/mksc

MARKETING SCIENCE
Vol. 43, No. 2, March–April 2024, pp. 267–279
ISSN 0732-2399 (print), ISSN 1526-548X (online)

Frontiers: A Simple Forward Difference-in-Differences Method

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Received: June 22, 2022
Revised: May 11, 2023; September 19, 2023
Accepted: October 19, 2023
Published Online in Articles in Advance: December 20, 2023

https://doi.org/10.1287/mksc.2022.0212

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Abstract. The difference-in-differences (DID) method is the most widely used tool to answer causal questions from quasiexperimental data in marketing and the broader social sciences. Because assignment to treatment in quasiexperiments is not random, the selection of proper control units is critically important for estimating the causal effect. DID requires that the treatment unit's outcomes would have been parallel to the average of the control units' outcomes in the absence of treatment. However, this DID parallel trends assumption is likely to be violated when assignment to the treatment and control groups is not random. We propose a simple forward difference-in-differences (Forward DID) method that uses a forward selection algorithm to flexibly select a relevant subset of control units. The Forward DID has several advantages. First, it can be widely applied and suitable even when DID is too restrictive. Second, Forward DID can accommodate any number of control units. Third, there are no overfitting concerns because Forward DID only needs to estimate one parameter after identifying a subset of control units. Fourth, Forward DID has computational advantages over algorithms that consider all possible subsets of control units. Finally, we establish consistency and develop inference theory, which is applicable to both stationary and nonstationary data. We demonstrate the usefulness of the Forward DID method and compare it with the alternative methods using simulations and an application to store openings.

History: Olivier Toubia served as the senior editor. This paper was accepted through the *Marketing Science*: Frontiers review process.
Supplemental Material: The online appendix and data files are available at https://doi.org/10.1287/mksc.2022.0212.

Keywords: causal effects • quasi-experimental methods • forward difference-in-differences • inference • average treatment effects on the treated • retailing

1. Introduction

Marketing researchers are interested in establishing causal effects in a variety of contexts (Busse et al. 2006, Goldfarb and Tucker 2011, Puranam et al. 2017, Guo et al. 2020, Berman and Israeli 2022). The difference-in-differences (DID) method is the most popular method to estimate treatment effects, such as the average treatment effect (ATE) and the average treatment effect on the treated (ATT), when using quasiexperimental data. DID is simple, easy to implement, and widely applicable. In fact, it even applies to data with only two time periods (one pretreatment period and one posttreatment period) and any number of treatment and control units. Marketing scholars have used DID to study how television advertising influences online shopping (Liaukonyte et al. 2015), how showroom openings affect sales (Bell et al. 2018), and the impact of a tax on sweetened beverages (Seiler et al. 2021) among many others (Mantin and Rubin 2016, Calzada and Gil 2020, Johnson et al. 2020, Todri 2022, Wang et al. 2022, Li and Van den Bulte 2023).

However, DID has a restrictive identifying assumption that may not be met in many data settings. The DID parallel trends assumption states that the treatment unit's outcomes would have been parallel to the average of the control units' outcomes during the posttreatment periods in the absence of treatment. Because this assumption is a statement about a counterfactual outcome of what the treatment outcomes would have been without treatment, it cannot be checked empirically. Therefore, a common practice is to check whether the parallel trends assumption holds during the pretreatment periods (which we call parallel pretrends assumption). Having parallel pretrends together with the assumption that the correlation between the treatment and control units' outcomes is stable during the posttreatment periods in the absence of treatment identifies the treatment effect (e.g., ATT). Therefore, selecting the most relevant control units so that the DID parallel pretrends assumption is likely to hold is of utmost importance. When selection into treatment is based on some observables, various covariate-matching methods, such as propensity score or *K*-nearest-neighbor matching, can be used to select the control units. When selection is based on unobservables, the extant literature does not offer a satisfying solution for how to select control units for the DID method.

However, there have been many recent developments expanding the causal inference tool kit beyond DID for marketers to include more flexible estimators, although they require long time series. Perhaps the most popular alternative approach is the synthetic control (SC) method proposed by Abadie and Gardeazabal (2003) and Abadie et al. (2010). The SC method relaxes the DID equal weight assumption and uses a weighted average of controls to estimate the treatment unit's counterfactual outcomes. Allowing for flexible weights can better handle heterogeneity between treatment and control units. The SC method has been used to estimate the effect of TV advertising on online chatter (Tirunillai and Tellis 2017) and the effect of banning foreign pharmacies from sponsored search (Chesnes et al. 2017). Another method proposed by Hsiao et al. (2012), which we will refer to as the HCW method, is an even more flexible method that imposes no constraints on the weights for controls units. However, these flexible methods also have limitations. First, the inference theory for the synthetic control method and HCW methods do not exist under certain data settings (e.g., nonstationary data where the structure of nonstationary is unknown). Second, the HCW method is not feasible when the number of control units is greater than the number of pretreatment time periods and likely suffers from overfitting.

Recently, Shi and Huang (2023) proposed the Forward HCW (FHCW) method, which uses a forward selection algorithm to select an appropriate subset of control units and then, applies the HCW method. Forward selection is an iterative algorithm that chooses a relevant subset of control units. It proceeds by first choosing the best subset of size 1. Then, it chooses the best subset of size 2 that includes the first subset. The algorithm continues to choose greater subsets that have one more control unit than the previous subset in the same way. Forward HCW can be applied even when HCW breaks down but has several limitations. First, similar to the synthetic control and HCW methods, there is no inference theory for nonstationary data of unknown structure, which is a key drawback as marketing data often fall into this category. Second, Forward HCW may still suffer from overfitting if it selects a large number of control units, and it is infeasible when the number of selected control units is larger than the number of pretreatment time periods.

We propose a simple forward difference-in-differences (Forward DID) method that takes advantage of the forward selection algorithm to select a relevant subset of control units and avoids the overfitting problem of the Forward HCW method. Forward DID flexibly selects the appropriate control units and therefore, is more likely able to accurately estimate the counterfactual and ATT than the DID method. Forward DID has several advantages. First, as a flexible alternative to DID, Forward DID can be widely applied and suitable even when DID is too restrictive. Second, Forward DID can accommodate any number of control units. Third, there are no overfitting concerns because Forward DID only needs to estimate one parameter after identifying a subset of control units using the forward selection algorithm. Fourth, by using a greedy algorithm that is a myopic approximation of an optimal algorithm, the Forward DID has computational advantages over algorithms that consider all possible subsets of control units. Fifth, we develop the inference theory, which is applicable to both stationary and nonstationary data. Finally, we establish the consistency of the Forward DID method. We compare the performance of the Forward DID method with alternative methods using simulations and an empirical application to store openings.[1]

2. Forward Difference-in-Differences Method

2.1. Notation

We first present the notation used for the DID method, which we will build upon for the Forward DID method. We consider the case of one treatment unit and N control units and extend to multiple treatment units in Online Appendix C. Let $\mathcal{N}_{co}$ denote the subset of control units that is used to estimate the ATT by the DID method, and let N_{co} denote the number of control units in $\mathcal{N}_{co}$.[2] The total number of time periods is T, so $t = 1, \ldots, T$. The outcome of the treatment unit at time t is $y_{tr,t}$, and the outcome of the ith control unit at time t is y_{it} for $i = 1, \ldots, N$. The treatment unit receives a treatment at time $T_1 + 1$ ($1 < T_1 < T$), and the N units that do not receive any treatment serve as potential control units.

We use $y^1_{tr,t}$ and $y^0_{tr,t}$ to denote the outcomes of the treatment unit with and without treatment, respectively. We use $y_{tr,t}$ without superscript to denote the observed outcome of the treatment unit at time t. Therefore, the treatment effect at time t for $t > T_1$ is $y^1_{tr,t} - y^0_{tr,t} = y_{tr,t} - y^0_{tr,t}$ because $y^1_{tr,t} = y_{tr,t}$ for $t > T_1$. The ATT is ($T_2 = T - T_1$ is the number of posttreatment time periods)

$$ATT = \frac{1}{T_2} \sum_{t=T_1+1}^{T} (y_{tr,t} - y^0_{tr,t}). \tag{2.1}$$

Because we only have one treatment unit, the average is over the number of posttreatment periods, T_2.

Because the counterfactual $y^0_{tr,t}$ for $t > T_1$ is not observable, we need to estimate $y^0_{tr,t}$ using observed data. Let $\overline{y}_{\mathcal{N}_{co},t} = N_{co}^{-1} \sum_{i \in \mathcal{N}_{co}} y_{it}$ be the average outcome of the selected control group. The identification of ATT is based on the following model specification: $y^0_{tr,t} = \alpha + \overline{y}_{\mathcal{N}_{co},t} + v_t$ or equivalently,

$$y^0_{tr,t} - \overline{y}_{\mathcal{N}_{co},t} = \alpha + v_t, \quad t = 1, \ldots, T, \tag{2.2}$$

where α is an unknown parameter (the intercept) and v_t is a zero mean, stationary idiosyncratic error term.[3]

We allow $y^0_{tr,t}$ and $\overline{y}_{\mathcal{N}_{co},t}$ to be nonstationary processes (e.g., nonlinear trend processes) of unknown structure provided v_t has zero mean and a central limit theorem applies to its partial sum. Model (2.2) implies that the DID parallel trends assumption holds as the possible nonlinear trending components cancel from $y^0_{tr,t}$ and $\overline{y}_{\mathcal{N}_{co},t}$, rendering the difference between the two processes a trendless stationary process. As we will show, model (2.2) identifies ATT and allows us to consistently estimate the ATT.

The ATT estimate can be obtained by first estimating model (2.2) using the pretreatment data

$$y_{tr,t} - \overline{y}_{\mathcal{N}_{co},t} = \alpha + v_t, \qquad t = 1, \ldots, T_1. \tag{2.3}$$

We use the least squares method to estimate α:

$$\hat{\alpha}_{\mathcal{N}_{co}} = \frac{1}{T_1}\sum_{t=1}^{T_1}(y_{tr,t} - \overline{y}_{\mathcal{N}_{co},t}). \tag{2.4}$$

Therefore, the estimated in-sample fitted value (for $t \le T_1$) and the out-of-sample predicted counterfactual (for $t > T_1$) are

$$\hat{y}^0_{tr,t} = \hat{\alpha}_{\mathcal{N}_{co}} + \overline{y}_{\mathcal{N}_{co},t}, \qquad t = 1, \ldots, T. \tag{2.5}$$

The estimated treatment effect at time t (with $t > T_1$) is $y_{tr,t} - \hat{y}^0_{tr,t}$. Therefore, the ATT is estimated by

$$\widehat{ATT} = \frac{1}{T_2}\sum_{t=T_1+1}^{T}(y_{tr,t} - \hat{y}^0_{tr,t}). \tag{2.6}$$

The advantage of considering a regression-based ATT estimate is that it allows us to easily analyze the quality of the fit by examining the goodness-of-fit R^2 of the regression model (2.3):

$$R^2_{\mathcal{N}_{co}} = 1 - \frac{\sum_{t=1}^{T_1}\hat{v}^2_{\mathcal{N}_{co},t}}{\sum_{t=1}^{T_1}(y_{tr,t} - \overline{y}_{tr})^2}, \tag{2.7}$$

where $\hat{v}_{\mathcal{N}_{co},t} = y_{tr,t} - \overline{y}_{\mathcal{N}_{co},t} - \hat{\alpha}_{\mathcal{N}_{co}}$ is the estimated residual from (2.3) and $\overline{y}_{tr} = T_1^{-1}\sum_{t=1}^{T_1} y_{tr,t}$ is the average of the treatment unit's outcome over the pretreatment period. We explicitly include $\mathcal{N}_{co}$ as a subscript in R^2 to emphasize that the quality of the fit depends on the subset of selected control units, $\mathcal{N}_{co}$. We turn to the problem of how to the select control units next in Section 2.2.

2.2. Forward Difference-in-Differences Algorithm

When using DID to estimate the treatment effect (e.g., ATT), researchers need to decide which control units to choose from the potential control units. Deciding how to select relevant control units is a challenging and important problem. When selection is based on observables, various covariate-matching methods have been proposed in the literature. See Abadie (2021) for a survey of various matching methods. The assumption behind the covariate-matching method is that when covariates of a control unit and a treatment unit are close to each other, then it is likely that their outcomes are parallel in the absence of treatment. However, this assumption can fail in practice. The similarity of outcomes may depend on some unobserved latent factors, and therefore, matching based on observed covariates may not be enough to ensure that DID can be used.

We propose a simple Forward DID method that first uses a forward selection algorithm to select a relevant subset of control units and then applies the DID method. Forward DID flexibly selects the appropriate control units and therefore, is more likely able to accurately estimate the counterfactual and ATT than the DID method. Next, we outline the steps of the Forward DID method.

Step 1. Select one control unit. Without loss of generality, choose the ith unit from the control pool. We then estimate ATT using Equations (2.3)–(2.7) but replace $y_{\mathcal{N}_{co},t}$ with y_{it} ($N_{co} = 1$ in this initial step). Do this for each of the N control units ($i = 1, \ldots, N$) to create N one-control-unit DID models. Choose the model with the highest R^2_i, where R^2_i is defined as in (2.7) but replace $\hat{v}_{\mathcal{N}_{co},t}$ with $\hat{v}_{it} = y_{tr,t} - y_{it} - \hat{\alpha}_i$, where $\hat{\alpha}_i = T_1^{-1}\sum_{t=1}^{T_1}(y_{tr,t} - y_{it})$. Therefore, this first step selects one submodel with $N_{co} = 1$.

Step 2. After selecting the first control unit using Step 1, choose another (the second) control unit from the remaining $N-1$ control pool. Do this one at a time, and add it to the one-control-unit model selected in Step 1. This results in $N-1$ two-control-unit models. Choose the model with the highest R^2 among these $N-1$ two-control-unit models.

Step 3. Continue the process by adding one more control unit, one at a time, to the previously selected model. At the end, we obtain a model containing all the N control units and therefore, have N selected submodels. The first one has one control unit, the second one has two control units, … , and the last one has N control units. Choose the model with the highest R^2 value from these N models. This is the model that the Forward DID method uses to estimate the ATT. We use $\widehat{\mathcal{N}}_{co}$ to denote the subset of controls selected by the Forward DID method.

Note that there is no concern of overfitting no matter how large N_{co} is as all models only contain a single unknown parameter α. Unlike a conventional regression model, all these submodels contain only one parameter α, and therefore, R^2 is identical to the adjusted R^2. The forward selection algorithm is a myopic approximation of the algorithm that would optimally sequence control units. However, allowing some degree of suboptimally buys us computational efficiency. This is especially important when the number of control units, N, is large and the optimal algorithm becomes intractable. Forward

DID is a viable and valuable alternative. To illustrate this, consider a setting similar to our empirical application with a pool of $n = 60$ control units. Forward DID only evaluates $1 + 2 + \ldots + 60 = 1,830$ submodels as opposed to $2^{60} > 1.15 \times 10^{18}$ for the optimal algorithm.

We would like to mention related recent literature on detecting latent group structure in panel data models (Bonhomme and Manresa 2015, Ando and Bai 2016, Su et al. 2016). It may be possible to use this approach to select control units whose sample path is likely to be parallel with the treatment unit. Specifically, assuming that a heterogeneous panel consists of a few homogeneous, latent subgroups, this recent literature can be used to detect and assign each control unit to a group. Then, in a similar spirit to the Forward DID method, the subgroup of control units with the highest correlation to the treatment unit during the pretreatment period is most likely to satisfy the parallel pretrends assumption and can be used to estimate a causal effect. However, theoretically justifying such an approach is beyond the scope of the current paper.

2.3. Inference Theory of the Forward DID Method

In this section, we present the identifying assumption of the Forward DID method before presenting the inference theory.

Assumption 2.1. *The subset of control units selected by the Forward DID method, $\widehat{\mathcal{N}}_{co}$, satisfies $y^0_{tr,t} = \alpha + \overline{y}_{\widehat{\mathcal{N}}_{co},t} + v_t$ for $t = 1, \ldots, T$, where α is a constant, $\overline{y}_{\widehat{\mathcal{N}}_{co},t} = \widehat{N}_{co}^{-1} \sum_{i \in \widehat{\mathcal{N}}_{co}} y_{it}$, $\widehat{N}_{co}$ is the number of control units in $\widehat{\mathcal{N}}_{co}$, and v_t is a weakly dependent process with zero mean and finite variance.*

Assumption 2.1 implies that the correlation between $y^0_{tr,t}$ and $\overline{y}_{\widehat{\mathcal{N}}_{co},t}$ is stable during the pretreatment and posttreatment periods in the absence of treatment. Assumption 2.1 is a Forward DID version of the parallel trends assumption, which we need to consistently estimate the ATT. Assumption 2.1 allows for $y^0_{tr,t}$ and $\overline{y}_{\widehat{\mathcal{N}}_{co},t}$ to be nonstationary processes of unknown form provided that the difference of the two processes is a stationary process. To better understand this assumption, we describe a situation where this assumption is violated. If an intercept-adjusted treatment unit is outside the range of the control units (e.g., the treatment's outcome has an upward trend that is steeper than that of all the control units' outcomes), then this assumption will be violated because no subset of control units can trace the steeper upward trend of the treatment unit. In such a case, the Forward DID method should not be applied, and researchers should consider alternative methods, such as factor model-based methods, modified synthetic control methods, or the augmented difference-in-differences method, which allow the treatment unit to be outside the range of the control units (Bai 2003, Doudchenko and Imbens 2016, Xu 2017, Li 2020, Bai and Ng 2021, Li and Sonnier 2023, Li and Van den Bulte 2023).

Next, we present the inference theory for the Forward DID method in the following proposition.

Proposition 2.1. *Under Assumption 2.1 and Assumptions 2–4 in Online Appendix A, as $T_1, T_2 \to \infty$,*

$$|\Pr(\sqrt{T_2}(\widehat{ATT}_{\widehat{\mathcal{N}}_{co}} - ATT)/\hat{\sigma}_{\widehat{\mathcal{N}}_{co}} \le a) - \Phi(a)| \to 0 \text{ for all } a \in \mathbb{R}, \tag{2.8}$$

where $\hat{\sigma}^2_{\widehat{\mathcal{N}}_{co}} = T_1^{-1} \sum_{t=1}^{T_1} \hat{v}^2_{\widehat{\mathcal{N}}_{co}}$, $\hat{v}_{\widehat{\mathcal{N}}_{co}} = y_{tr,t} - \overline{y}_{\widehat{\mathcal{N}}_{co},t} - \hat{\alpha}_{\widehat{\mathcal{N}}_{co}}$, and $\Phi(\cdot)$ is the standard normal cumulative distribution function.

The proof of Proposition 2.1 is in Online Appendix B.

Our inference theory for the Forward DID method accommodates both stationary and nonstationary data of known and unknown structure. In addition, it is simple and straightforward to apply as it is based on normal inference theory. In fact, the availability of widely applicable inference theory for the Forward DID method is one of its major advantages compared with other flexible alternatives to the DID method, such as the synthetic control method, the synthetic control method with an intercept, the HCW method, and the Forward HCW method. There is no inference theory for the synthetic control method (with and without an intercept), the HCW method, and the Forward HCW method when the data are nonstationary of unknown structure. Unlike the synthetic control method (with and without an intercept), which uses a weighted average of control units, or the HCW and Forward HCW methods, which use a linear combination of the control units, the Forward DID method, like DID, uses a simple, unweighted average of the control units and therefore, avoids estimating a large number of parameters. It is this simplicity that lends itself to straightforward inference theory. In the appendix, we present a practitioner's guide outlining the steps of using the Forward DID method to estimate the ATT and conduct inference.

To examine the finite sample performance, we use simulations to compare the Forward DID method with alternative methods (DID, SC, synthetic control with an intercept (SC_I), and Forward HCW).[4] The full simulation results are in Online Appendix E. We find that the Forward HCW method likely overfits, resulting in large mean square errors (MSEs) for all cases. For data-generating processes (DGPs) where DID is applicable, DID and SC methods perform best as expected, although Forward DID still performs well. For DGPs where DID is not applicable, Forward DID and the synthetic control method with an intercept (Abadie and L'hour 2021) perform best, with the Forward DID outperforming the synthetic control method with an

intercept when the pre- and posttreatment periods are shorter.

2.4. Consistency of the Forward DID Method

In this section, we establish consistency of the Forward DID method. We interchangeably use consistency of the Forward DID method and consistency of the forward selection algorithm used by the Forward DID method. We can see from (2.7) that selecting a submodel with the largest R^2 is equivalent to selecting a model with the smallest sum of squared residuals ($SSR = \sum_{t=1}^{T_1} \hat{v}^2_{\mathcal{N}_{co},t}$) or equivalently, to minimizing the empirical error variance defined by $T_1^{-1}\sum_{t=1}^{T_1} \hat{v}^2_{\mathcal{N}_{co},t}$. Therefore, we can reinterpret the forward selection algorithm (used by the Forward DID method) as minimizing the empirical variance in each step (as it is equivalent to maximizing R^2).

Because a submodel selected by a forward selection algorithm may be different from an optimal submodel chosen after examining all the possible 2^N submodels, we use a similar forward selection algorithm under the assumption that the true variances are known as the benchmark. The theoretical forward selection algorithm selects a submodel in exactly the same way as the empirical forward selection algorithm (used by the Forward DID method), except that it replaces the empirical variance by the true variance. That is, in each step of the theoretical forward selection algorithm, we compare true error variances of different submodels and select a submodel that has the smallest true error variance: $E(v^2_{\mathcal{N}_{co},t}) = E[(y_{tr,t} - \overline{y}_{\mathcal{N}_{co},t} - \alpha_{\mathcal{N}_{co}})^2]$.

We present the consistency result of the forward selection algorithm used by the Forward DID method in the following proposition.

Proposition 2.2. *Assume that the theoretical forward selection algorithm only selects a single subset, $\mathcal{N}^*_{co}$, and N is a fixed positive integer. Then, under Assumption 2.1 and Assumptions 2–4 in Online Appendix A,*

$$\Pr(\widehat{\mathcal{N}}_{co} = \mathcal{N}^*_{co}) \to 1 \text{ as } T_1 \to \infty.$$

The proof of Proposition 2.2 is in Online Appendix D. Proposition 2.2 shows that when N is fixed and the theoretical forward selection algorithm selects a single subset $\mathcal{N}^*_{co}$, then with probability approaching one (as T_1 increases), our forward selection algorithm used by the Forward DID method will select the same subset as the infeasible theoretical forward selection algorithm. That is, the Forward DID method can consistently select the relevant subset of control units.

The final set selected by the theoretical forward selection algorithm may not be a singleton because there may be ties during the selection process. Establishing consistency while allowing for ties requires more complicated notation and more sophisticated proofs, which we deal with in Proposition D.1 in Online Appendix D. In Online Appendix D, we also relax the fixed N assumption and prove consistency for the case that allows N to increase with T_1 (Proposition D.2 in Online Appendix D). Therefore, the consistency results presented in Proposition 2.2 and Online Appendix D (Propositions D.1 and D.2) are one of the main theoretical contributions of the paper in addition to the inference theory presented in Section 2.3, and it grounds the Forward DID method with another good theoretical property.

3. Store Opening on Sales

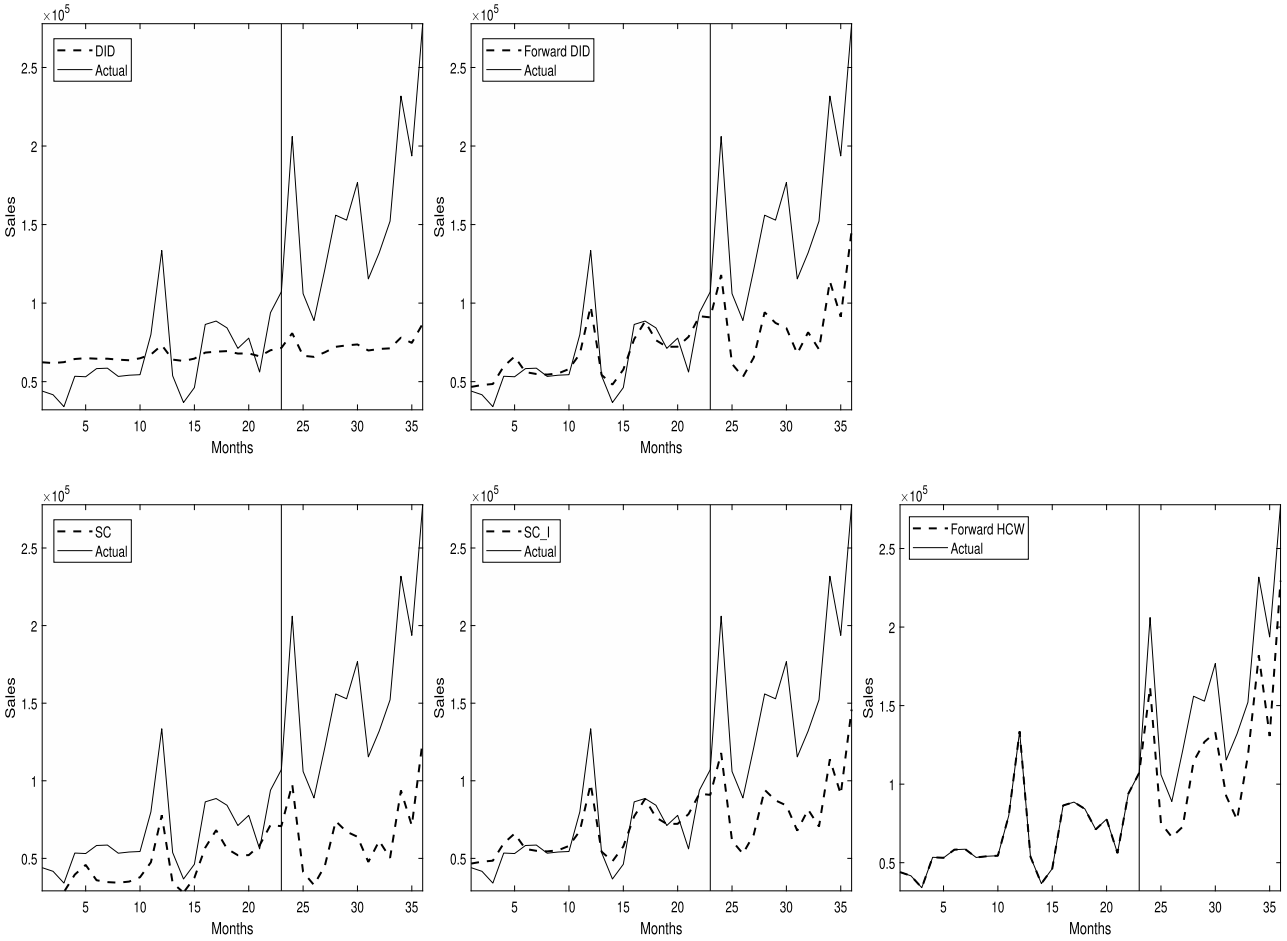
Changing consumer needs have made it imperative for online retailers to identify factors that are critical for its growth. We study an online-first retailer that sells men's clothing and accessories. When the retailer launched in 2007, it started completely online and later opened brick-and-mortar stores in certain cities so that customers could touch and try the products before deciding to purchase. However, given the high cost of opening a physical store, it is important to accurately assess the benefit or the additional increase in sales because of the store opening.

We estimate the causal effect of opening a store in a city on the retailer's sales in the city. We use the Forward DID method and compare it with four alternative methods: (1) DID, (2) SC, (3) synthetic control with an intercept (SC_I), and (4) Forward HCW. We cannot consider the HCW method because it is not feasible in these empirical applications, where the number of control units (60) is larger than the number of pretreatment time periods (23 or 24). We have data from January 2014 to December 2016. During this time period, the retailer opened a store in Atlanta, Georgia in December 2014 and opened stores in San Diego, California and San Jose, California in January 2015. These were the first stores to open in their respective cities of Atlanta, San Diego, and San Jose. Therefore, these 3 cities are the treatment units, and 60 cities without stores serve as the potential control units. The DID method uses all 60 control units, whereas the other methods (SC, SC_I, Forward HCW, and Forward DID) generally use a subset of the control units.

3.1. ATTs and Confidence Intervals

A necessary condition for a method to be applicable is that the sales estimated by that method need to fit the actual sales well in the pretreatment period. To check this, we plot the sales in Atlanta, San Diego, and San Jose along with the estimated in-sample fit/counterfactual using Forward DID and alternative methods in Figures 1–3. In each of the three figures, the solid lines are the actual sales, and the dashed lines are the in-sample fit in the pretreatment period ($t \le T_1$) and the out-of-sample predicted counterfactual sales in the posttreatment period ($t > T_1$). For instance, for the Forward DID method, the

Figure 1. Atlanta Sales and Fitted/Predicted Curves by Method



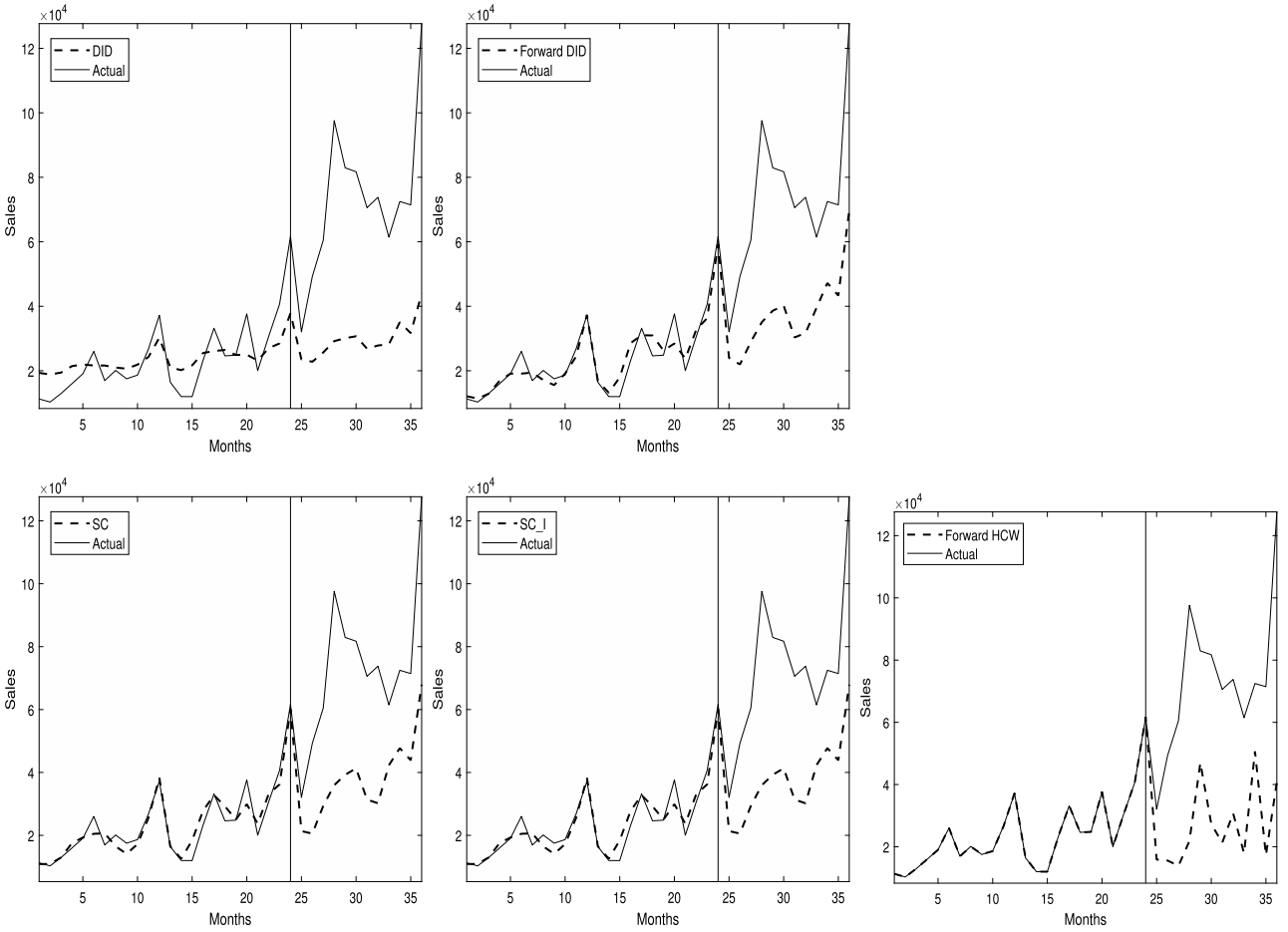
Notes. (Upper left panel) DID method. (Upper right panel) Forward DID method. (Lower left panel) SC method. (Lower center panel) SC_I method. (Lower right panel) Forward HCW method.

in-sample fit in the pretreatment period is computed via (2.5) (i.e., $\hat{y}^0_{tr,t} = \hat{\alpha}_{\mathcal{N}_{co}} + \overline{y}_{\mathcal{N}_{co},t}$ for $t = 1, \ldots, T_1$). The out-of-sample predicted counterfactual sales in the posttreatment periods are also computed via (2.5) but with t belonging to a posttreatment period ($t = T_1 + 1, \ldots, T$). The pretreatment patterns in these figures provide insight into which methods are applicable and which are not (and therefore, should be excluded) for the Atlanta, San Diego, and San Jose cases.

Figure 1 shows the results for Atlanta. We can clearly see that DID (upper left panel) and SC (lower left panel) do not fit the pretreatment data well, whereas the Forward DID (upper right panel), SC_I (lower center panel), and Forward HCW (lower right panel) do. Therefore, for Atlanta, DID and SC are not applicable and should be not be used. DID uses an average of all 60 control cities' sales, which has a much flatter trend compared with Atlanta's steep upward trend during the pretreatment period. Therefore, the DID counterfactual in the posttreatment period is likely underestimated, which results in overestimating the ATT. SC also does not fit the Atlanta data well in the pretreatment period. Specifically, the SC fitted curve is below the actual sales. Because the SC method does not allow an intercept, the SC fitted curve does not have the flexibility to shift upward to match Atlanta's pretreatment sales. Therefore, similar to the DID, the SC method underestimates the counterfactual, which results in overestimating the ATT. By contrast, the Forward DID, SC_I, and Forward HCW methods fit the in-sample data well in the pretreatment period. However, the Forward HCW method likely overfits the in-sample data because of having to estimate many parameters. In Section 3.2, we compare the out-of-sample performance by method and confirm that indeed Forward HCW performs poorly out-of-sample. Therefore, for Atlanta, DID, SC, and Forward HCW should be not be used, whereas Forward DID and SC_I remain as applicable methods.

Figure 2 shows that for San Diego, the DID fits the pretreatment data poorly, whereas the other four methods (SC, SC_I, Forward HCW, and Forward DID) fit the in-sample data well. Similar to Atlanta, the DID method

Figure 2. San Diego Sales and Fitted/Predicted Curves by Method

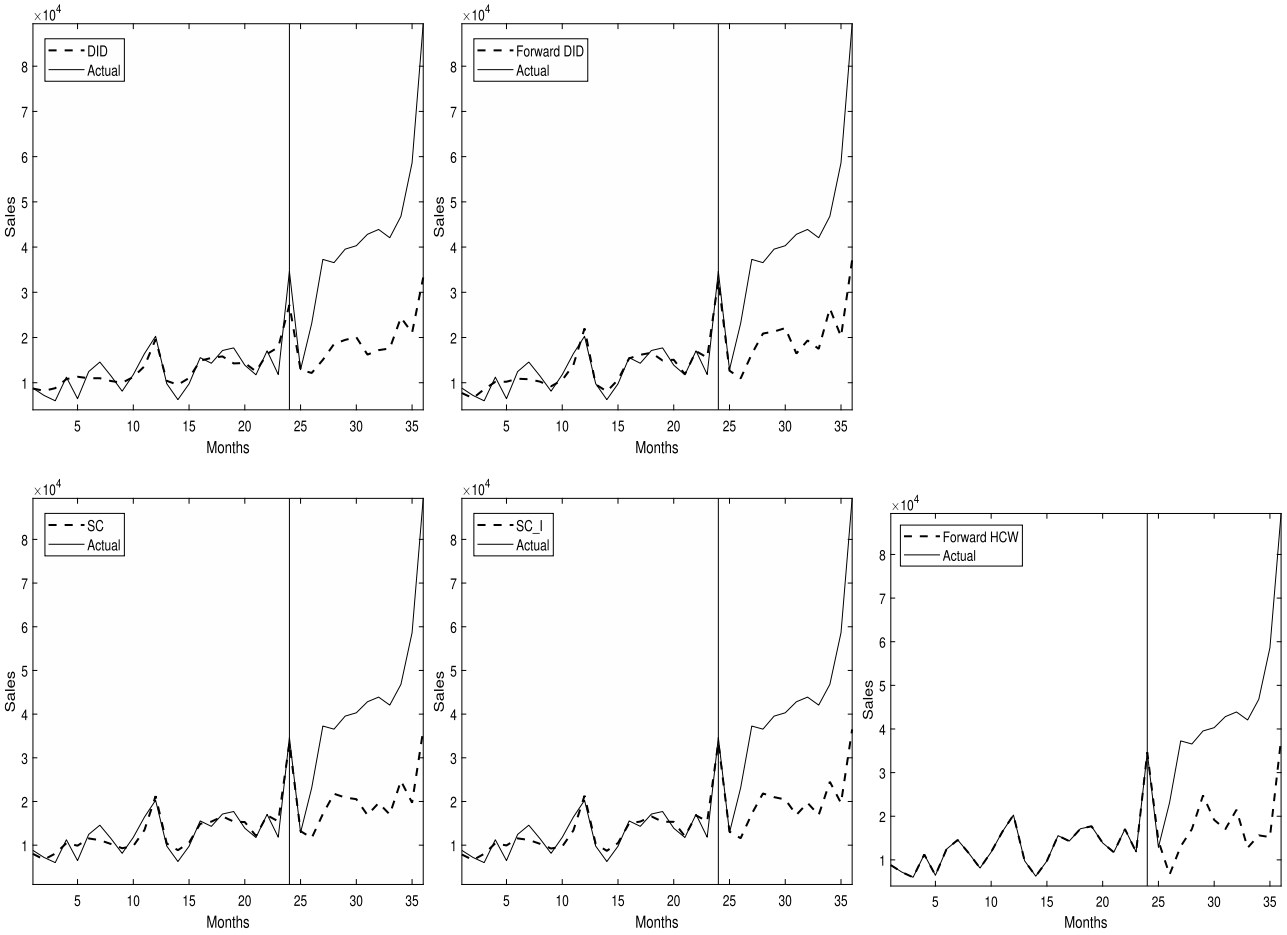


Notes. (Upper left panel) DID method. (Upper right panel) Forward DID method. (Lower left panel) SC method. (Lower center panel) SC_I method. (Lower right panel) Forward HCW method.

fits the in-sample data poorly because of its inability to match the upward trends of San Diego's actual sales. As a result, DID likely underestimates the counterfactual in the posttreatment and overestimates the ATT. By contrast, the Forward DID, SC, SC_I, and Forward HCW methods fit the pretreatment data well. Finally, Figure 3 shows that for San Jose, all five methods fit San Jose's pretreatment sales well. Similar to the Atlanta case, even though the Forward HCW method fits the in-sample data well for San Diego and San Jose, it likely suffers from overfitting because of having to estimate many parameters. In Section 3.2, we compare the out-of-sample performance by method and confirm that indeed Forward HCW performs poorly out-of-sample. Therefore, for San Diego, DID and Forward HCW should not be used, whereas Forward DID, SC, and SC_I remain as applicable methods. For San Jose, Forward HCW should be excluded, whereas the other four methods remain as applicable methods. Although we have discussed which methods should be excluded for each city, for the sake of completeness, we report the ATTs and confidence intervals (CIs) for all methods in this section and the prediction mean square errors (PMSEs) for all methods in Section 3.2.

Next, we compare the ATT and goodness of fit. Table 1 shows the ATT, ATT percentage, and R^2 by method for each of the three cities. Before we discuss the results by city, first note that for the Forward HCW method, the R^2 values are greater than 0.99 for all three cities, which is another indication of overfitting. The number of pretreatment time periods is 23 for Atlanta and 24 for San Diego and San Jose. When using Forward HCW, we need to restrict the maximum number of selected control units to be 21 so that the maximum number of parameters is 22 (maximum number of control units plus one for the intercept), which is less than the number of pretreatment time periods. In each of the three empirical applications, Forward HCW selected the maximum-allowed 21 control units. In other words, Forward HCW runs a regression with 22 regressors to fit 23 or 24 data points (pretreatment time periods). Therefore, the Forward HCW suffers from overfitting the in-sample data. By contrast, other methods

Figure 3. San Jose Sales and Fitted/Predicted Curves by Method



Notes. (Upper left panel) DID method. (Upper right panel) Forward DID method. (Lower left panel) SC method. (Lower center panel) SC_I method. (Lower right panel) Forward HCW method.

(DID, SC, SC_I, and Forward DID) have proper regularization conditions to prevent overfitting. Specifically, the Forward DID has implicit regularization and only estimates one parameter after identifying a subset of control units; therefore, overfitting is not an issue. Therefore, we exclude Forward HCW because its ATT is not reliable when discussing results in Table 1, which we do so next.

Next, we discuss the ATT, ATT percentage, and R^2 results by city in Table 1. Recall from the discussion of Figure 1 that for Atlanta, DID and SC are not applicable because of their poor in-sample fit and likely overestimate the ATT. This is consistent with the results in Table 1 that show R^2 for DID and SC being very low compared with the other methods and the ATTs from DID and SC being much greater than the ATT from Forward DID and SC_I. By contrast, Forward DID and SC_I fit the pretreatment data well and have an R^2 of 0.76. Forward DID and SC_I suggest that opening a store in Atlanta led to an increase in monthly sales of \$75,143 (86%). Recall from the discussion of Figure 2 that for San Diego, DID is not applicable because of its poor in-sample fit, and it likely overestimates the ATT. This is consistent with the results in Table 1 that show for San Diego that the SC, SC_I, and Forward DID methods

Table 1. ATT and ATT Percentage by City and Method

	ATT					ATT percentage					R^2				
	DID	SC	SC_I	F_{DID}	F_{HCW}	DID	SC	SC_I	F_{DID}	F_{HCW}	DID	SC	SC_I	F_{DID}	F_{HCW}
Atlanta	89,144	95,314	75,143	75,143	45,208	121	142	86	86	39	0.23	0.05	0.76	0.76	0.99
San Diego	43,840	35,861	35,859	35,913	47,136	148	95	95	96	179	0.55	0.92	0.92	0.89	1.00
San Jose	23,782	22,814	22,831	22,665	24,874	125	114	114	113	132	0.78	0.91	0.91	0.90	1.00

Table 2. 95% CI and CI Width Ratio for Each City by Method

	95% CI			CI width ratio (over Forward DID)		
Method\city	Atlanta	San Diego	San Jose	Atlanta	San Diego	San Jose
DID	[74,882, 103,402]	[38,405, 49,274]	[21,875, 25,690]	1.80	2.16	1.43
SC	[87,264, 101,423]	[31,681, 41,376]	[18,836, 25,447]	0.858	1.89	2.48
SC_I	[29,314, 115,262]	[15,564, 58,424]	[6,726, 36,844]	5.24	8.53	11.3
Forward DID	[67,212, 83,074]	[33,399, 38,426]	[21,331, 23,998]	1.00	1.00	1.00
FHCW	[45,207.9, 45,208.1]	[47,135.9, 47,136.0]	[24,873.9, 24,874.1]	2.5×10^{-6}	3.7×10^{-7}	9.7×10^{-5}

have much larger R^2 compared with DID, indicating that they are likely more reliable in this setting and that the ATTs estimated using SC, SC_I, and Forward DID are smaller than the ATT from DID. Opening a store in San Diego led to a monthly increase in sales in the range of \$35,859–\$35,913 (95%–96%) according to SC_I, SC, and Forward DID. Finally, for San Jose, DID, Forward DID, SC, and SC_I methods have high R^2 and similar ATTs ranging from \$22,665 to \$23,782 (we exclude Forward HCW because of overfitting). As we can see from Table 1, in each city, the different methods that are applicable for that city produce similar ATT estimates, which is a good sanity check.

One major advantage of Forward DID over other flexible alternative estimators to DID is inference theory that is widely applicable. We develop the inference theory for the Forward DID method, which is applicable to both stationary and nonstationary data of known and unknown structure. Like DID, Forward DID uses a simple (unweighted) average of control units, which avoids having to estimate a large number of parameters and therefore, leads to simple inference. By contrast, other alternative methods to DID, such as the synthetic control methods (SC and SC_I) that use a weighted averaged of control units and the HCW and Forward HCW methods that use a linear combination of control units, do not have inference theory when data are nonstationary of unknown structure.

We now compute the CIs for the Forward DID and alternative methods (DID, SC, SC_I, and Forward HCW). Although we compute the confidence intervals for the SC, SC_I, and Forward HCW methods for the sake of completeness, the confidence intervals for these methods may not be reliable for the following reasons. For the SC and SC_I methods, there is no inference theory when the data are nonstationary of unknown structure, which is the setting of this empirical application. To compute the CIs for SC and SC_I, we apply the subsampling procedure proposed by Li (2020), whose validity requires that the number of pretreatment time periods is much larger than the number of control units and that the data are stationary data or nonstationary data of known structure. The Forward HCW method as discussed previously suffers from overfitting in-sample, which means that the ATT is unreliable. In addition, the near-perfect in-sample fit results in the estimated error variance being virtually zero and the CI almost shrinking to a single point at $\widehat{ATT}_{FHCW}$. However, we cannot interpret these narrow CIs of the Forward HCW as more precise estimates but rather, interpret them as the method breaking down because of overfitting. Therefore, when discussing the CIs and CI width ratios in Table 2, we exclude Forward HCW.

Table 2 shows the 95% CIs and CI width ratios. The CI width ratio is defined as the width of the 95% CI of an alternative method over the width of the 95% CI of the Forward DID method. Therefore, a ratio greater than one indicates that the alternative method has a wider CI than the Forward DID method. There is only one ratio (we exclude Forward HCW) that is smaller than one, which corresponds to Atlanta using SC, but recall that SC is not applicable for Atlanta. All other ratios (excluding Forward HCW) are greater than one, which indicates that Forward DID has narrower confidence intervals compared with alternative methods.

Finally, notice that for this store opening application, Forward DID and SC_I are the only methods (of those considered) that are applicable in all three cities, and they perform very similarly in terms of in-sample fit and ATT estimates. Therefore, we take this opportunity to discuss in general when Forward DID is preferable to SC_I and when it is not. Although Forward DID and SC_I lead to similar ATTs when they are both applicable, Forward DID is preferable to SC_I when Forward DID is applicable (i.e., its parallel trends assumption holds) because inference theory is more widely available for Forward DID, which leads to more reliable confidence intervals. Specifically, inference theory is available for Forward DID for stationary and nonstationary data of known and unknown structure, whereas for SC_I, inference theory does not exist when data are nonstationary and the structure of the nonstationary is unknown. By contrast, SC_I is preferable to Forward DID when Forward DID is not applicable, but SC_I is still applicable. This can occur because SC_I is more general than Forward DID as SC_I allows weights on different control units to be different.

Therefore, in this store opening application, even though Forward DID and SC_I perform similarly in terms of in-sample fit and ATT estimates, Forward DID

is preferable to SC_I because the availability of widely applicable inference theory results in more reliable confidence intervals. In addition, Forward DID's confidence intervals are more narrow compared with SC_I's confidence intervals, which means we get tighter, more informative bounds and are able to better quantify the uncertainty using Forward DID compared with SC_I. In the next section, we show that Forward DID also performs better overall out-of-sample compared with SC_I.

3.2. Out-of-Sample Prediction MSE

In this section, we use a backdating exercise to compare the out-of-sample prediction by method. For the backdating exercise, we only use the pretreatment data ($t \leq T_1$) and split it into two parts. We pretend that there was an intervention at T_0+1, where $T_0 < T_1 - 1$, so that the first T_0 time periods are the new "pretreatment" period and the last $T_1 - T_0$ time periods are the new "posttreatment" time period.

We then fit the model using new "pretreatment" data $1 \leq t \leq T_0$ and predict counterfactual outcome $y^0_{tr,t}$ for $T_0+1 \leq t \leq T_1$. Let $\hat{y}^0_{tr,t}$ denote the counterfactual estimate of $y^0_{tr,t}$. Because there is no intervention for $t \leq T_1$, we actually observe $y^0_{tr,t}$ for $t \leq T_1$. Therefore, we can compute PMSE by

$$\text{PMSE} = \frac{1}{T_1 - T_0} \sum_{t=T_0+1}^{T_1} (y_{tr,t} - \hat{y}^0_{tr,t})^2, \qquad (3.1)$$

where $\hat{y}^0_{tr,t}$ is one of the five predictors using the DID, Forward DID, SC, SC_I, and Forward HCW methods.

Table 3 shows the PMSE for the Forward DID method (divided by 10^7 for readability). To more easily compare Forward DID's PMSE with alternative methods, Table 4 shows the PMSE ratios of a given method (DID, SC, SC_I, and Forward HCW) over Forward DID. Therefore, PMSE ratios greater than one in Table 4 indicate that Forward DID performs better out-of-sample. Note that using results from Tables 3 and 4, we can obtain the PMSEs for all methods.

Now, we discuss the results by method starting with Forward HCW. Recall that Forward HCW likely overfits the pretreatment data because of estimating 22 parameters to fit 23 or 24 data points (pretreatment time periods) and having very high R^2 for in-sample fit. Therefore, we expect Forward HCW to perform poorly out-of-sample, which is indeed the case as indicated by large PMSE ratios with one exception for all three cities in Table 4.

Table 3. $PMSE/10^7$ for the Forward DID Method

T_0	Atlanta	San Diego	San Jose
12	11.126	2.2248	1.3842
14	12.147	2.2658	1.5641
16	12.541	3.5044	1.9491
18	16.099	4.2309	2.5349
20	25.857	1.6783	3.0483

Next, we discuss the DID method. When DID fits well in-sample (San Jose) (see Figure 3), it performs the best out-of-sample as indicated by having the smallest PMSE ratios. However, when DID does not fit well in-sample (Atlanta and San Diego) (see Figures 1 and 2), it performs poorly out-of-sample as indicated by large PMSE ratios. Therefore, one takeaway is that DID should be used when it is applicable, and only when DID's identifying assumption is likely violated should we then consider more flexible alternatives, such as Forward DID and synthetic control methods. Next, we discuss the synthetic control methods without (SC) and with an intercept (SC_I). When SC does not fit well in-sample (Atlanta) (see Figure 1), it performs poorly out-of-sample as indicated by large PMSE ratios. In this case, adding an intercept to SC improves its performance. For Atlanta, SC_I performs similarly to the Forward DID as indicated by PMSE ratios equal to one (four of five cases) or close to one (one of five cases). In the two other cases (San Diego and San Jose) where SC fits well in-sample (see Figures 2 and 3), SC and SC_I perform similarly and are both outperformed by Forward DID.

Finally, we discuss the Forward DID method. From a bird's-eye view, 43 of 60 PMSE ratios are greater than one, and 4 of 60 PMSE ratios are equal to one, which indicates that Forward DID performs very well out-of-sample

Table 4. PMSE Ratio by Method

	City											
	Atlanta				San Diego				San Jose			
T_0/ratio	$\frac{DID}{FDID}$	$\frac{SC}{FDID}$	$\frac{SC_I}{FDID}$	$\frac{FHCW}{FDID}$	$\frac{DID}{FDID}$	$\frac{SC}{FDID}$	$\frac{SC_I}{FDID}$	$\frac{FHCW}{FDID}$	$\frac{DID}{FDID}$	$\frac{SC}{FDID}$	$\frac{SC_I}{FDID}$	$\frac{FHCW}{FDID}$
12	2.28	2.36	0.992	0.461	3.14	1.03	1.07	3.99	0.358	0.817	1.22	2.12
14	3.08	7.85	1.00	5.26	2.59	0.977	0.987	3.12	0.705	1.36	1.41	3.54
16	4.41	9.51	1.00	2.79	3.16	1.41	1.16	9.50	0.468	1.47	1.39	3.25
18	2.26	7.57	1.00	3.90	2.59	1.39	1.11	9.11	0.428	1.17	1.28	4.82
20	1.60	5.50	1.00	1.60	2.54	0.798	2.14	3.66	0.453	0.740	0.975	1.94

Note. FDID, forward difference-in-differences.

relative to other methods. Delving deeper into the 13 cases where Forward DID is outperformed, 5 of the cases are when DID performs best for San Jose. In this case, Forward DID comes in second after DID and thus, performs best among the alternatives to DID. Of the remaining eight cases where Forward DID is outperformed, Forward DID still outperforms alternative methods overall. In other words, for a given column that contains one of the remaining eight cases (where the ratio is less than one), at least three or four of the five scenarios with different T_0 favor Forward DID. In summary, when DID is applicable, it should be used as it outperforms alternative flexible methods. However, Forward DID performs best out of the alternative flexible methods. When DID is not applicable, Forward DID performs favorably in the vast majority of cases in out-of-sample prediction among alternative methods.

In summary, this store entry setting is a good place to use Forward DID (relative to other methods) because Forward DID is applicable in all three cities (i.e., parallel pretrends assumption holds), it has a simple and widely applicable inference theory to calculate confidence intervals, it does not suffer from overfitting, and it performs well out-of-sample.

4. Conclusion

The difference-in-differences method is the most popular quasiexperimental approach for estimating causal effects. It is simple, easy to implement, and applicable for any number of treatment and control units and for any number of pretreatment and posttreatment time periods. However, its identifying parallel trends assumption may be violated in quasiexperimental data where the treatment and control units are heterogeneous. There have been many recent developments expanding the causal inference tool kit to include more flexible estimators, such as the synthetic control method (with and without an intercept), the HCW method, and the Forward HCW method. However, these flexible methods also have limitations. First, their inference theory does not exist for nonstationary data of unknown structure. Second, the (Forward) HCW is not feasible when the number of (selected) control units is greater than the number of pretreatment time periods and likely suffers from overfitting.

We propose a simple Forward DID method that takes advantage of the forward selection algorithm to flexibly select a relevant subset of control units. The Forward DID has several advantages. First, it can be widely applied and suitable even when DID is too restrictive. Second, Forward DID can accommodate any number of control units. Third, there are no overfitting concerns because Forward DID only needs to estimate one parameter after identifying a relevant subset of control units. Fourth, Forward DID has computational advantages over algorithms that consider all possible subsets of control units. Fifth, we develop the inference theory that is applicable to both stationary and nonstationary data of known and unknown structures. Finally, we establish the consistency of the Forward DID method. The inference theory and the consistency results are the main theoretical results of the paper. To facilitate ease of use, we provide a practitioner's guide outlining steps of how to use Forward DID to estimate the ATT and conduct inference in the appendix. Although we only consider estimation of ATT based on treatment and control units' outcome variables, future research could consider adding relevant covariates to the Forward DID method.

We calculate the ATT and compare the out-of-sample performance of the Forward DID method with four alternative methods (DID, SC, SC_I, and Forward HCW) using simulations (presented in Online Appendix E) and an empirical application to store openings. In simulations, DID and SC methods perform best for DGPs where DID is applicable, which is expected, although Forward DID still performs well. For DGPs where DID is not applicable, Forward DID and the synthetic control method with an intercept perform best, with the Forward DID outperforming the synthetic control method with an intercept when the pre- and posttreatment periods are shorter. In the empirical applications, the Forward DID method performs well in all cases and outperforms alternative methods in the out-of-sample prediction. Forward DID and SC_I are the only methods (of those considered) that are applicable in all three cities, and although they perform very similarly in terms of in-sample fit and ATT estimates, Forward DID is preferable to SC_I because inference theory is more widely available, which leads to more reliable confidence intervals.

As with any methodology, it is also important to understand its boundary conditions. The Forward DID method should not be used when its identifying parallel trends assumption is violated. An example of this scenario would be when the intercept-adjusted treatment is outside the range of the control units (e.g., the treatment unit's outcome has an upward trend that is steeper than that of all the control units' outcomes). In such a case, it is not possible to find a subset of control units whose average traces the treatment unit well in the pretreatment period.

Fortunately, because of the flexibility of the forward selection algorithm, there are many empirical marketing and business settings in which the Forward DID is applicable. The Forward DID belongs to a class of flexible alternatives to the DID method. These flexible alternative methods should be used when DID is not applicable because of its identifying parallel trends assumption being violated. Compared with other flexible alternatives to the DID method (e.g., SC with or without an intercept, HCW, and Forward HCW), the Forward DID

method is especially useful to researchers in settings where inference is important and data are nonstationary of unknown structure. Whereas the synthetic control method (with and without an intercept), the HCW method, and the Forward HCW method lack widely applicable inference theory, the Forward DID method has a simple normal inference theory, which we derive, that is widely applicable to both stationary and nonstationary data of known and unknown structures. The Forward DID method is a powerful causal inference tool that complements existing methods well and expands the set of causal research questions that researchers can answer.

Acknowledgments

The author is grateful for insightful comments from the editor, the associate editor, and two reviewers that greatly improved the paper. The author also thanks Unnati Narang, Garrett P. Sonnier, Raji Srinivasan, Ioannis Stamatopoulos, and Christophe Van den Bulte; the participants of the 2022 Artificial Intelligence, Machine Learning and Business Analytics Conference, the 2023 Cornell Young Scholar Camp, and the McCombs School of Business Research Colloquium; and the marketing seminar participants at the University of California Riverside School of Business, the University of Florida Warrington College of Business, and Carnegie Melon University Tepper School of Business for valuable comments. The author certifies that there are no affiliations with or involvement in any organization or entity with any financial interest or nonfinancial interest in the subject matter or materials discussed in this manuscript.

Appendix. A Practitioner's Guide

In this appendix, we outline the steps of using the Forward DID method to estimate ATT and conduct inference.

Step 1. Use the Forward DID forward selection algorithm described in Section 2.2 to select a subset of controls that has the largest R^2. Let $\widehat{\mathcal{N}}_{co}$ denote the selected subset.

Step 2. Compute the counterfactual outcomes using (2.5) (i.e., $\hat{y}^0_{tr,t} = \hat{\alpha}_{\hat{\mathcal{N}}_{co}} + \overline{y}_{\hat{\mathcal{N}}_{co},t}$, where $\hat{\alpha}_{\hat{\mathcal{N}}_{co}}$ is given by (2.4), $\overline{y}_{\hat{\mathcal{N}}_{co},t} = \widehat{N}_{co}^{-1}\sum_{i\in\hat{\mathcal{N}}_{co}} y_{it}$, and $\widehat{N}_{co}$ is the number of control units in $\widehat{\mathcal{N}}_{co}$).

Step 3. Plot $\hat{y}^0_{tr,t}$ and $y_{tr,t}$ for $t = 1,\ldots,T_1$, and check whether parallel pretrends between $\hat{y}^0_{tr,t}$ and $y_{tr,t}$ hold (i.e., whether $\hat{y}^0_{tr,t}$ traces $y_{tr,t}$ well in the pretreatment period). Visual inspection is an effective way of checking this.

Move to Step 4 if the parallel pretrends hold. Otherwise, do not use the Forward DID method, and consider other methods whose parallel pretrends hold.

Step 4. Estimate ATT by $\widehat{ATT} = \frac{1}{T_2}\sum_{t=T_1+1}^{T}(y_{tr,t} - \hat{y}^0_{tr,t})$, where $\hat{y}^0_{tr,t}$ is computed in Step 2.

Step 5. Compute the $(1-\alpha)$ confidence interval of ATT by

$$\left[\widehat{ATT} - c_{1-\alpha/2}\hat{\sigma}_{\mathcal{N}_{co}}/\sqrt{T_2},\ \widehat{ATT} + c_{1-\alpha/2}\hat{\sigma}_{\mathcal{N}_{co}}/\sqrt{T_2}\right],$$

where c_α is the upper αth quantile of a standard normal random variable. For example, the 95% confidence interval of ATT is given by

$$\left[\widehat{ATT} - 1.96\hat{\sigma}_{\mathcal{N}_{co}}/\sqrt{T_2},\ \widehat{ATT} + 1.96\hat{\sigma}_{\mathcal{N}_{co}}/\sqrt{T_2}\right].$$

Endnotes

[1] MATLAB and R code implementing Forward DID using a publicly available data set can be found on the *Marketing Science* website (following the journal's replication policy) and on the author's website.

[2] In mathematical terminology, N_{co} is the cardinality of the set $\mathcal{N}_{co}$.

[3] Both α and v_t also depend on $\mathcal{N}_{co}$. However, for notational simplicity, we often do not include $\mathcal{N}_{co}$ as a subscript for α and v_t.

[4] We do not consider the HCW method in simulations because it is not feasible when the number of control units is larger than the number of pretreatment time periods.

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